

Kota Buku AI classroom programme

The complete team sourcebook

Research, decisions, product experience and delivery reasoning.

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Sifututor leads the proposed product and technology delivery, powered by Learnest Lab. Kota Buku's programme and support responsibilities are our proposed arrangement, not its confirmed acceptance. Nothing here is a deployment authorisation, final quotation or legal opinion.

How this book works

The six reading parts are: opportunity (chapter 1), reasoning (2–3), experience (4–5), how it works (6–9), delivery (10–14), and maintaining the truth (15–16 and appendices). Chapters retain the approved I01–I16 order. A new team member should read 1–5 and 10–12 first; engineers deepen through 6–9; commercial and governance reviewers use 3, 9, 13 and 15; presentation writers use 1, 4, 5 and 16.

Sources are cited by title and section. Original research is dated 6 September 2026; reconciliation and authoring are dated 7 September. A historical source assessment is not a fresh verification of today's law, service, curriculum or product capability. The evidence register records which primary sources were reopened. Material unresolved facts remain unresolved.

1. Understand the opportunity, people and proposal

In plain language: we are proposing a teaching assistant that helps teachers understand each student and the class, choose suitable support and follow up. The personal touch comes from the teacher; the proposed system makes learning evidence easier to use and helps prepare the chosen response. Generating material serves this purpose, rather than defining the product. This connects preparation, classroom work, evaluation and appropriate family involvement; it is not an existing app under a new name. Core evidence summaries, class views and rules support richer AI assistance in a same-launch guided trial with selected teachers, as explained in chapter 5.

Hafiz's brief combines teacher-workload relief with an ambitious classroom vision. If a teacher saves time preparing an exercise but must spend more time correcting it or copying records elsewhere, the product may not help. Conversely, a reliable learning record can support a teaching action that the teacher wanted to take but previously lacked time to prepare. Both efficiency and educational usefulness matter.

Party or reader	Proposed contribution or need
Kota Buku	Programme engagement, school/training coordination and outward-facing delivery; proposed first-line enquiries, subject to agreement.
Sifututor / Learnest Lab	Product direction, design, development, operation, onboarding material, trainer guidance and technical escalation. Sifututor fronts/contracts; Learnest Lab remains visible.
Teachers	A useful daily workflow, trustworthy material, control over decisions and accessible support.
Students and parents	Suitable learning support and curated own-record information, not unrestricted monitoring.
Wider programme parties	Device/connectivity/distribution arrangements outside our software commitment; exact interfaces and owners need confirmation.

Kota Buku's own published remit includes book-related digital infrastructure and access. Connecting permitted content to classroom activities is our interpretation of an opportunity, not an official appointment or an established classroom-product mandate. [Kota Buku About page](#), checked 7 September 2026.

The deck builds interest. The external document explains the proposal. A separate reconciled commercial offer supports an interested party's decision. This sourcebook equips the team behind all three; it is not automatically safe to share externally. The partner's internal founder briefing is not a recipient-ready commercial offer. [Reconciliation §§1–3](#).

2. Understand what the research taught us

The research question was broader than “which AI model?” We needed to understand teacher tasks, curriculum structure, existing record systems, comparable education products, hardware limits, adoption and governance. The useful result is a set of design implications with evidence limits, not a list of impressive statistics.

Research area	Lesson carried into the proposal	What the evidence does not establish
Teacher preparation	Guided AI drafting is worth evaluating as part of a complete teacher task.	A universal time saving or this product's effectiveness.
Existing school records	Add useful learning-process evidence; avoid recreating sensitive administrative datasets.	That no internal government learning records or APIs exist.
Curriculum	Preserve subject, year, standard and version so evidence remains interpretable over time.	That a curriculum label certifies an AI answer or makes scores comparable.
Classroom assessment	Structured responses provide a more checkable capture path than unrestricted handwriting.	That every paper response is captured correctly or every student has a device.
National programmes	Examine training, connectivity, adoption and accountability alongside hardware.	A universal single cause for failure, or proof that one vendor structure guarantees success.
Family portals	Curated information and deliberate permissions fit the approved direction.	A scientifically prescribed age threshold for every school context.

The EEF/NFER December 2024 evaluation studied guided ChatGPT lesson preparation in secondary-school science in England. It found reduced preparation time and no detected difference in sampled resource quality. This supports testing the workflow, not transferring its result to Malaysian teachers, all subjects or student attainment. [Original report](#), [executive summary](#), reopened 7 September 2026.

The earlier research examined DIKSHA/Sunbird and Kolibri for content/offline architecture, Uruguay for sustained programme support, and several national device programmes for delivery difficulties. Those are dated research cases, not a controlled comparison proving what causes success. We borrow questions and documented patterns, not a claim that their results transfer. Their exact scale, legal outcomes and current status are not reasserted here. [Technical research §§2.1/12](#).

Local and global product comparisons also serve different purposes. A vendor webpage can show how an offering is described, while an independent evaluation can test a defined outcome. Neither establishes our implementation readiness. Parent-portal patterns, teacher-generation tools and assessment platforms informed the six pillars; their marketing adoption counts, compliance claims and headline gains should not become ours. [Research §§4–6/8/10](#).

The communication research points toward benefits, concrete examples and distinct readiness stages. Hafiz settled the editorial choice: explain the experience first, then Kota Buku's opportunity and delivery. Prior-programme problems and policy alignment support the case rather than lead the pitch. [Communication references](#); REC-03.

3. Understand why these decisions were made

Owner approval answers what we intend to offer. Evidence answers what is true. Counterpart acceptance answers what others have agreed to do. These are separate columns in our reasoning; merging them creates false commitments.

Decision	Approved direction and reason	Important boundary
D1 / D9	Co-branded Kota Buku app, delivered by Sifututor, powered by Learnest Lab. Combines programme identity and our product responsibility.	Exact legal entity and IP/processing schedules unresolved; no automatic joint venture.
D2	M1-or-later recommended; A16 minimum supported target. Explain genuine trade-offs.	Both require product validation. Deterministic marking is not M1-exclusive; device prices/delta belong to partner costing.
D3	Keep concept work local-only.	No commit or un-ignore; a fresh clone does not include this package.
D4	One grounding backend, staged exposure: template planning and AI practice for all at launch; full AI-RPH guided beta for selected teachers at the same launch.	Exposure controls risk without deferring the entire full-plan backend; quality is not presumed.
D5 / D10	Partner owns commercial figures; setup/run-fee structure remains context.	No duplicated costing, affordability or acceptance of estimates inferred.
D6	Kota Buku/KPM determine mandatory versus optional framing.	Do not substitute a “never mandate” recommendation; staged feature exposure remains.
D7	Comprehensive purpose-bound learning capture and governance; externally describe reporting and analytics.	No indiscriminate collection or headline national-data-asset pitch.
D8	Supplied devices are teachers-only.	Parent/student own-device web access remains; younger students still benefit without accounts.
D11 · 7 September 2026	Class views, rule-based flags/suggestions, notice/reminder templates and parent digest for all teachers at launch; plain-language AI analysis, teacher assistant and AI-written admin drafts in a guided trial with selected teachers at that same launch.	Deck stands; earlier Fast-follow placement is superseded. Readiness, D7 processing and teacher review must be validated. Full-RPH beta under D4 remains.
C4	Blend workload reduction and the six-pillar AI-classroom vision.	Neither intended benefit is a measured outcome.

Alternatives were genuinely considered. Deferring all AI lesson planning would reduce exposure but lose the launch vision; D4 instead pairs a universal template route with guided full-plan use. Building an autonomous learning path would remove teacher steps but also remove necessary interpretation; the chosen model keeps the teacher responsible. Copying another app's code would suggest speed, but the dated audit found structural mismatches with native/offline school workflows. These are reasons, not excuses to narrow the approved offer. [Handoff §7](#) and [§13](#); [Product §§1/5/6](#).

The partner framework contributes useful discipline: source records, responsibilities, milestones, challenges and safe editions. We did not adopt automatic research overrides, arbitrary defect quotas, mandatory seven-document production, hidden sibling branding, assumed official integration or automatic AI cuts. The reconciled package covers those useful subjects within the reader paths and the separate commercial stage. [Framework review](#); REC-01–REC-07.

4. Understand the whole teaching and family journey

Use this fictional fractions lesson with Cikgu Farah and Aiman Hakim in 4 Bestari to understand the relationships. It is not an approved curriculum example or a real student record. A lesson objective links preparation to the evidence collected afterwards; it does not itself validate a question or grade.

Moment / actor	Input and action	Output and human decision
Teacher prepares	Select subject/year/standard; use a template; request draft practice.	Teacher-approved plan and questions, with source/version and approval state.
Teacher teaches	Use downloaded material, a projected activity or paper exercise.	Planned versus actual activity can be confirmed; showing a slide is not evidence every student answered.
Student responds	Supported digital response or a labelled bubble/tick sheet.	An observable answer; response time exists only if the actual channel measures it.
Teacher checks	Review uncertain captures, identity match and answer key.	Confirmed response/score and any override reason; missing responses remain missing.
Teacher follows up	Examine evidence and a traceable rule-based suggestion.	Reteach, adapt, gather further evidence or take no action; record the chosen response.
Teacher shares	Select progress highlights and approve a periodic digest.	Parent/eligible student sees permitted own-record information, not all working notes.
Next lesson	Review new evidence and prior intervention.	Refine the next plan; do not infer causal improvement merely because scores changed.

The fallback matters at every step. If generation fails, the template and approved library remain usable. If a scan is unclear, the teacher corrects or re-enters the response rather than accepting a guessed score. If the connection fails, locally supported work must show whether it is saved or pending synchronisation. If family connectivity is unavailable, classroom teaching must not depend on portal use. [Product §§3–9](#); [Data §5 measurement boundary](#).

A later photo cannot reconstruct how long the student originally thought, what hints they used or whether they discussed an answer. These must not be invented as events. A mark is also different from a mastery estimate, which is different again from a teacher-confirmed formal TP judgement. Keeping those distinctions is the basis of trustworthy recommendations.

5. Understand the six-pillar product atlas

Core means proposed launch scope, not finished software. **Fast-follow** is the next stage subject to its prerequisites. **Later** requires further maturity or validation. The full-RPH, AI-analysis, teacher-assistant and AI-admin guided trial is a selected-teacher same-launch exposure rule, not a synonym for Fast-follow. Appendix A indexes every feature row; the profiles below explain why each pillar exists. [Product §§2–13](#).

Pillar 1: learning evidence

The teacher needs more than an overall score. Formative check-ins, structured marking and teacher-maintained mastery records connect responses to the learning objective. A standalone attendance register provides operational context but is not evidence of mastery. Inputs must be matched to the correct student and assessment version; the output is reviewable evidence, not an automated verdict. Handwriting and official-system sync are not Core shortcuts. Acceptance needs capture checks, teacher override and a clear missing-data state.

Pillar 2: classroom activity

The teacher should be able to use material during the lesson, not only prepare it beforehand. Core polls/quizzes and downloaded resources support that purpose. The classroom route can use paper, a shared display or available student devices; it must state what is actually observed. Networked individual responses require connectivity and permitted access. Simple gamification and textbook-linked activities are Fast-follow, not reasons to assume behavioural tracking or content rights.

Pillar 3: content preparation

Reusable RPH structures and AI practice address repeated preparation. Reading-level adaptation helps the teacher prepare a more suitable explanation. The teacher selects the objective and checks generated content, including answer key, language and difficulty. Full AI-RPH uses the same grounding service but starts with selected teachers in guided beta at launch. Four product languages are a Core requirement, not proof of equally good generated output. Textbook-grounded materials depend on rights/access; student-specific material depends on reliable learning evidence.

Pillar 4: teaching support and personal attention

The central experience is **understand** → **support** → **follow up**, not “generate more content”. Help the teacher evaluate observed learning evidence for each student, recognise shared needs and decide what to try. The personal touch belongs to the teacher: a conversation, a different explanation, targeted practice or a class discussion. Content generation serves that decision. This is an intended benefit, not measured effectiveness.

Core student summaries and rule-based suggestions make the next action explainable. A suggestion should point to the observations that triggered it, and the teacher can reject it. These are not all generative AI functions. Core class views and rule-based flags/suggestions support every teacher at launch. Plain-language AI analysis, teacher assistant conversations and AI-written administrative drafts deepen this support in a guided trial with selected teachers at that same launch (D11, Hafiz, 2026-09-07). Templates and the curated parent digest remain Core. For example, shared difficulties could inform the teacher's choice of a temporary small-group explanation or class reteaching—not automatic grouping or fixed ability labels. After support, the teacher reviews new evidence without assuming it proves causation. Missing data must remain visible. Autonomous assignment without teacher choice is not the chosen model; a formal TP decision remains with the teacher. [Product §6](#).

Pillar 5: everyday efficiency

Communication templates and shared lesson plans reduce repeated drafting and support reuse. Approval matters: a parent summary is not automatically sent simply because a draft exists. Official attendance integration remains permission-dependent, and handwriting capture remains validation-dependent. Wellbeing/behaviour tools are not a headline promise or permission to collect sensitive family information. Their appearance as a Later research option does not override the data exclusion rules.

Pillar 6: family access

Parents and appropriately aged students need understandable progress rather than a raw continuous feed. Core contains curated progress, attendance/announcements, age controls and periodic visibility on a separate web surface. Direct messaging is Fast-follow. Exact age bands and permissions, particularly the middle years, remain open; do not turn illustrative age examples into approved policy. No student devices are supplied. Acceptance must test own-child isolation, permission changes and understandable summaries as well as successful sign-in.

The intended scope remains Standard 1–Form 5 and government/private/international contexts. A rollout may require validated combinations of subject, year, language and curriculum, but no author may silently narrow the approved scope. Conversely, broad scope is not a claim that every combination is already validated. The coverage matrix must distinguish intended coverage, source availability, content review and actual release readiness.

6. Understand what exists and what must be built

The dated 6 September internal codebase review found useful operating patterns in Kelasapp and LLS, not a ready-made native school application. Kelasapp's attendance/identity patterns and LLS's controlled parent views, teacher-led services and AI-output validation can inform design. Their existence does not prove code portability or shorten every differentiating workstream. [Product §1](#).

Layer	Intended responsibility	Validation needed
Teacher iPad	Local workflow, supported capture, resource cache, saved/pending status.	Target devices, accessibility, offline behaviour and data protection.
School service	School/class/student relationships, permissions and authoritative records.	Correct scoping, transfers and identity; not a tuition billing schema copied wholesale.
Curriculum/content service	Source/version identifiers, permitted resource retrieval and material provenance.	Taxonomy, permissions, publication/withdrawal behaviour and language coverage.
Generation service	Retrieve suitable source context, generate structured drafts and reject invalid references.	Teacher-reviewed correctness, privacy, quality, failure handling and actual Malaysian processing.
Family web	Curated own-record access and appropriate notices.	Age/guardian policy, visibility and correction/complaint paths.
Operations	Logs, recovery, monitoring and support.	Measurable capacity, restore evidence, incident responsibility and restricted access.

Offline-first means useful work can be saved and completed locally where supported; it does not mean every feature works without internet. New cloud generation and network responses still need a connection. A pending update must not look delivered. Proposed implementation acceptance should test retried submissions without duplicate records, stale edits, interrupted capture and safe conflict resolution. Exact conflict policy remains an implementation decision to specify and test, not an already-implemented feature. [Technical §§2/5/8](#).

D2's M1-or-later recommendation and A16 supported minimum remain. Do not say ordinary marking or all local machine learning requires Apple Intelligence. That named product's eligibility and app-supplied local processing are different questions. This book does not add unverified RAM, speed or model-support claims. Device, fleet-account and distribution eligibility still need the existing checks; the proposed managed route is not permission to assume every account arrangement works. [Technical §§3/6/7 and corrections](#).

7. Understand curriculum, content and AI quality

Curriculum is the organising reference, not just a label added to a generated paragraph. A record needs the curriculum version, subject, year/form, content standard and learning standard. Otherwise a revised standard may silently change the meaning of older evidence. The proposed hierarchy takes a standards-exchange approach; it does not claim certified interoperability. [1EdTech CASE](#), checked 7 September 2026; [Technical §4](#).

The source pipeline is **permitted document** → **versioned content** → **relevant retrieval** → **structured draft** → **automated checks** → **teacher review** → **approved material**. Each step has a different job. Retrieval finds relevant material; validation rejects broken references or malformed output; the teacher decides whether the teaching content is suitable. Passing a format check is not passing a pedagogy review.

The Malaysian August 2025 preprint compared grounded and ungrounded Form 1 mathematics MCQ generation. Its corpus and automated evaluation were narrow, and it did not provide human validation. It motivates testing grounding, not claiming validated full RPH or all-curriculum accuracy. [Original paper §§3/6.3](#), checked 7 September 2026.

Source rights are a separate gate. D4 removes dependence on Kota Buku textbook access for the initial public-DSKP approach; it does not establish unrestricted rights to every public document. Textbook retrieval/generation requires permission for that processing, not merely permission to read an e-book. Record what may be stored, transformed, displayed and retained, with withdrawal handling. [Product §§5/10/13](#).

Quality review needs representative subject/year/language cases, answer verification, difficulty checks and teacher edit/reject evidence. Evaluate MCQ generation separately from full lesson plans. Capture failures also need separate measurement: accurate grading logic cannot compensate for misread handwriting. A foreign inference fallback is not authorised for identifiable work under the Malaysia-residency decision.

8. Understand data: what, why and who can see it

The principle is comprehensive learning capture within observable, approved purposes, not comprehensive surveillance of children. Every field needs an actor, source, purpose, access rule and retention decision. Automatic collection is not automatically justified. [Data §§5–8](#).

Record family	Examples and provenance	Key control
Lesson	Class/subject/version, intended objective, actual taught standard, resource and reflection.	Separate plan from teacher-confirmed activity; opening a resource is not proof of learning.
Item/attempt	Item/source version, answer, score, mode, teacher approval and override reason.	Time/hints only where actually observed; paper scans cannot reconstruct them.
Student learning	Evidence history, teacher-confirmed mastery, interventions and access state.	Estimate versus formal judgement distinct; missing evidence is not a zero.
Class/school	Coverage, permitted summaries and intervention load.	Stage availability and access restrictions; standard IDs do not equate scores.
Teacher workflow	Feature use, errors, draft acceptance/editing, time proxies.	Not a work appraisal or proof of the policy teaching/admin time split.
Family interaction	Notice/permission state, curated views and complaints.	Own-record access, data minimisation and age policy.

The proposed event envelope preserves event/message identity, time, actor, object, school/class/subject context, curriculum version, device/session/offline state and event data. It is informed by the previously reviewed Sunbird pattern, not a copied production schema. Stable event identity supports retry handling and audit; timestamps need the distinction between recorded time and later server receipt. The implementation must define those contracts rather than invent them from a presentation. [Technical §2.1](#); [Data §5](#).

Teacher G1/G2 access and student-record access are not the same rule. Individual teacher reporting is limited to that teacher and the immediate school leader; above-school teacher reporting is aggregated/anonymised. Student records have separate documented school/controller authorisation routes. Parents and students see only their permitted own-record view. Exceptional service/security access is minimised, logged and reviewed, not a reporting tier.

Do not collect household income, aid details, marital status, disability, health, religion/ethnicity, biometrics, psychometric results, location tracking or sensitive home-life notes merely because a source system has them. Even a contextual flag needs a permitted purpose. Do not feed teacher workflow data into appraisal or public rankings. Pseudonymised records remain potentially identifiable; aggregation needs suppression and linkage-risk checks.

Access is specific to a purpose

Reader	Student-record boundary
Subject/class teacher	Own students/classes.
Head teacher / GPK	Own school, for a documented duty such as intervention or moderation; individual access logged.
PPD / JPN	District/state aggregates; individual drill-down only under controller authorisation, logged.
KPM headquarters	National aggregates; individual access through a documented, authorised controller process, logged.
Parent / student	Only the permitted, age-appropriate own-record view.
Researcher	Approved pseudonymised extracts inside a controlled environment, not a general download right.
Public	Suppressed, rounded statistics; no public ranking of schools or classes.
Operational staff	Minimised exceptional service/security access, logged and reviewed.

These are proposed student-access rules, not permission to bypass the stricter teacher G1 rule. Research requires vetted/trained people, an approved purpose, a controlled no-download setting, minimised pseudonymised data and checked outputs—the “Five Safes”. No direct identifiers or free-text extracts; no re-identification. Secondary use needs its own approved basis and applicable consent. A contract and technical controls must prohibit independent vendor reuse, commercial exploitation, teacher appraisal/discipline/ranking and feeds to e-Prestasi/PBPPP. These are proposed safeguards, not signed terms or settled statutory authority. [Data §§7.1–7.8.](#)

The record has a lifecycle

A student record connects lesson context, observed item attempts, teacher-confirmed judgements and intervention history over time. A move to another school must use a controller-authorised transfer and documented portable format, not an unrestricted export or assumed KPM connector. Phase-1 generic export does not establish official format acceptance. Retention is set by the controller for each data class: the learning record is proposed to follow the student through Form 5 plus a defined period; raw prompts and workflow telemetry are short-lived. Exact periods remain open. Deletion follows controller instruction; end-of-service return/deletion and backup handling require an agreed schedule. Every individual read/export needs a tamper-evident audit trail visible to the controller, with its own retention decision. [Data §§5.3/7.6.](#)

AI quality and usage records also need purpose limits: prompt category, model/version and generation provenance, draft acceptance/edits, usage units and controlled cost-governance records are not a licence to retain raw prompts indefinitely. This sourcebook describes the record requirement, not commercial amounts or current model pricing. [Data §5.5.](#)

9. Understand safeguards and operating resilience

In plain language: someone must be accountable for permitted data use, and the software must enforce that account. Brand names do not settle legal roles. The intended controller/processor division requires exact entities, instructions, notices, retention and incident schedules before operation. [Data §7/§11](#).

Identifiable processing must stay in Malaysia, including generation, OCR, logs and backups. A provider offering Malaysian servers is not sufficient proof that every subservice stays there. Map the actual path and test it. Recovery also needs demonstrated restores, not just a backup checkbox. Support access should be exceptional and reviewable; exported data must follow the same restrictions as the screen.

Incident handling needs a named owner, detection/triage, containment, evidence preservation, escalation, relevant notification assessment, recovery and follow-up. Exact legal triggers/deadlines cannot be inferred from a generic checklist. The prior review corrected overstatements about automatic vendor exemption and procurement exposure. Preserve the qualified treatment in Technical §15 and Handoff §13; obtain current authoritative/legal applicability checks before any commitment. This sourcebook deliberately does not restate unverified deadlines, tax liability or commencement as current conclusions.

Safeguard	Evidence the team should obtain
Purpose and roles	Controller-approved purposes, processor instructions and named accountable entities.
Family notice/access	Age/guardian rules, understandable notices, correction and complaint journey.
Non-punitive teacher data	Contract prohibition plus permissions, exports and audit tests.
Residency	Actual service/data-flow evidence across inference, storage, logs and recovery.
Accessibility	Web WCAG 2.2 AA target and applicable native assistive-technology checks.
Resilience	Restore/retry tests, failure communication and support escalation evidence.

WCAG 2.2 is a target, not proof of conformance; native iPad behaviour needs its own tests. [W3C standard](#), checked during external production on 7 September 2026. None of these design requirements authorises collection, deployment or a legal conclusion on its own.

10. Understand onboarding and support

Adoption begins with completing useful work. A first-use journey should guide a teacher through attendance, one practice item, one checked response and a lesson plan. Guidance must explain what to review, not train teachers to accept every AI suggestion. Four-language materials and school champions are part of the proposed Core enablement approach. [Product §2.1](#); [Technical §8.1](#).

Work	Proposed lead	Handoff / accountability
Product onboarding and trainer guidance	Sifututor / Learnest Lab	Teaching tasks, known limits and technical help routes explained.
School coordination and training arrangements	Kota Buku	Participating context and logistical needs passed to the product/support team.
First-line enquiries	Kota Buku	Resolve known questions; escalate with safe, sufficient evidence.
Platform investigation/fixes	Sifututor / Learnest Lab	Diagnose, communicate workaround/status and confirm resolution.
Adoption/quality review	Joint	Review completed tasks, recurring problems and changes required.

This is REC-05's approved proposal, not agreed staff or service levels. Do not assume IPGM/IAB or another institution has committed training capacity. Device/connectivity faults need a separate programme support route; the software team must not promise to replace hardware outside its scope.

An escalation should describe the task, expected behaviour, what occurred, environment and a minimised example. Do not request unrestricted student records as a routine troubleshooting shortcut. Where a problem affects a lesson, communicate the useful fallback first, then the technical status. Closing a ticket should mean the teacher can complete the task, not merely that a developer changed code.

11. Understand delivery sequence and dependencies

The launch plan is substantial: Core across six pillars, with full AI-RPH, AI analysis, teacher assistant and AI-admin drafts in a selected-teacher guided trial at the same launch. Delivery workstreams must prepare for that experience together rather than finishing isolated components and calling the whole product ready. [Product §12](#).

Workstream	Evidence before release	If not ready
Teacher/family workflow	Representative end-to-end tasks, correct permissions and usable language/accessibility.	Fix the workflow; do not substitute screenshots for working evidence.
Curriculum/generation	Source permissions, taxonomy coverage and teacher-reviewed output.	Use approved templates/resources where viable; no unsupported generated claim.
Capture and records	Correct student/item pairing, uncertain-mark handling, teacher override and missing-data behaviour.	Manual confirmation or supported alternate capture, not silent guessing.
Offline/service operation	Saved/pending distinction, retry/conflict behaviour, capacity, residency and recovery.	Bound the supported workflow and resolve material launch requirements.
Enablement/support	First-use guidance, proposed-to-agreed owner transitions and staffed escalation.	Resolve responsibilities; no assumed training throughput.
Guided full-RPH / analysis / assistant / admin trial	Selected-teacher access, review process and feedback loop on shared backend.	Do not expose unvalidated full-plan, analysis, assistant or AI-admin trial functions universally.

Fast-follow includes automatic history-based individualised material, direct messaging and conditional content/capture/interface expansion. A prerequisite resolved in one lane does not automatically validate the others. Textbook rights do not prove teaching quality; an API permission does not prove compatibility; a model-quality result does not prove Malaysian processing.

There is no new fixed date, headcount or whole-school cohort rule in this book. D6 leaves mandatory/optional framing to the relevant authorities. Delivery acceptance and the partner's commercial milestone model must use matching definitions, but this sourcebook does not approve its payment schedule or impose new contractual gates.

12. Understand how usefulness will be measured

Ask whether the teacher could identify a need, choose a suitable individual or group response and review subsequent evidence—not merely produce another worksheet. Group-insight evaluation follows that feature's release stage; Core rule-based class views and selected-teacher AI interpretation need separate checks; neither is evidence of implemented capability.

Measure the work the product is meant to improve. A count of generated questions says nothing about whether they were correct, used or helpful. The proposed evaluation should combine workflow burden, teaching quality, adoption, family comprehension and operational reliability. These are proposed measures, not established baselines or promised targets.

Measure	Definition / collection approach	Misinterpretation to avoid
Whole-task time	Time from beginning preparation/checking/reporting to usable completion, including correction and re-entry.	Draft-generation latency as total teacher time saved.
Draft usefulness	Accepted, edited and rejected outputs with reasons, by task/language/curriculum.	Acceptance alone as correctness or student benefit.
Capture quality	Verified identity/item matching and unresolved/error rates in representative class sets.	OCR accuracy alone as reliable end-to-end marking.
Completed/repeated workflow	Successful real task and return use over an agreed window.	Login count as beneficial adoption.
Teaching follow-up	Teacher choice, evidence consulted and later observed work.	A score change as causal proof of the intervention.
Family comprehension	Whether the recipient understands progress, limits and next action.	Portal visits as understanding.
Support/reliability	Task-blocking incidents, recovery and recurring support demand.	A healthy landing page as a functioning teacher journey.

Start with representative baseline tasks, then compare the same task definitions during use. Record context such as year/subject/language, connectivity and first-time versus experienced use. Do not publish small identifiable groups; teacher monitoring restrictions still apply to evaluation. Agree thresholds, evaluators and escalation before claiming pass/fail. Independent evaluation must remain distinct from vendor testing if described as independent.

Learning outcomes require an appropriate evaluation design beyond operational metrics. Time saved, adoption and attainment are separate outcomes. A useful first result may be that the teacher completes preparation with less correction burden; it is not evidence of improved examination results. [Research §6.6.1/§6.7; REC-05.](#)

13. Understand the commercial interface and boundaries

The partner owns the numbers. Our job is to make the scope and assumptions underneath them intelligible and consistent. Neither a figure-free sourcebook nor an approved product direction validates the commercial model. The [partner founder briefing](#) and [read-only review](#) remain controlled internal sources, with later Handoff corrections taking precedence.

Product/delivery input	Commercial implication to reconcile
Teacher-only supplied devices; family own-device access	Use the correct population for each workload, licence and support assumption.
Core and same-launch selected-teacher AI trial	Price the shared backend and controlled exposure without silently deleting scope.
AI usage/caching/quality	Validate workload and current service economics; do not reuse stale model prices.
Malaysia processing and recovery	Include real compliant service paths and support burden, not just generic hosting.
Proposed support split	Name responsibilities, staffing assumptions and escalation before accepting service commitments.
Content/integration dependencies	Define inclusion, exclusions, change control and delay/acceptance treatment.

D1's positions include our platform/code ownership, explicit processing roles, no exclusivity beyond the KPM programme and retained ability to offer the Learnest Lab version to private/international schools. These are our negotiating positions, not executed rights against Kota Buku. Exact legal entity, inter-entity IP and processing schedules require Hafiz's confirmation. Do not borrow the TEKUN document's different platform-ownership model.

The future recipient-safe commercial offer must reconcile assumptions, legal applicability and disclosure before sharing. The internal founder pack includes challenge material that should not be copied wholesale. Restricted legal/security issues should be tracked by owner and practical dependency without reproducing raw sensitive detail in this whole-team book.

14. Understand future capability without hidden promises

The long-term value is in useful connections. Reliable lesson/response records can support better-informed practice choices; permitted content can support textbook-specific activities; verified interfaces can reduce duplicate entry. None of these unlocks all other features automatically.

Foundation	What it could enable	What still needs proof
Reliable learning evidence	Student-specific material and class-pattern insights.	Data quality, teaching validity and safe access.
Permitted textbook corpus	Chapter-linked prompts and exercises.	Processing rights, ingestion quality and teacher review.
Verified official interface	Compatible reporting or later synchronisation.	Current format, authorisation, failure handling and operating ownership.
Quality across languages/years	Broader release across intended curriculum contexts.	Each combination's coverage and acceptance, not translation alone.
Useful support/evaluation loop	Sustained improvements and responsible expansion.	Real capacity, recurring demand and evidence of usefulness.

Private/international contexts remain in the approved scope, but their governance, curriculum and operating models must be mapped separately. The existence of a Malaysian standards taxonomy does not make all international curricula equivalent. Retained commercial rights are not a validated go-to-market plan or permission to start outreach.

Externally, describe reporting and analytics. Internally, understand the value of longitudinal learning evidence while applying the same purpose and access constraints. A more complete dataset is not automatically a representative or causally interpretable dataset. Expansion must preserve teacher authority and the family's understandable experience.

15. Understand assumptions, contradictions and open checks

An assumption is useful when its consequence is visible. It becomes dangerous when a later writer mistakes it for agreement or implementation. Track source/date, evidence class, approval, counterpart acceptance, impact if false, owner and verification route separately.

Open item / source	If false or unresolved	Responsible next evidence route
Partner role/capacity, REC-02/05	Proposed support and training cannot become service commitments.	Hafiz/partner agreement; no agent outreach.
Legal identity, controller/IP schedules, H D1/D9	Contract/data accountability remains incomplete.	Hafiz and appropriate legal/contract review.
Source rights, P §§5/10/13	Public access does not permit every processing use.	Exact terms and rights-holder evidence.
Curriculum/year/language matrix, P §13	Broad scope cannot be called validated coverage.	Curriculum/product quality review within existing scope.
Middle-year age gates, P §8	Core access cannot use invented permissions.	Governance and pedagogical policy decision.
Paper capture, P §9.1	Misread/mispaired records could drive wrong advice.	Representative classroom worksheet evidence with appropriate permissions.
Malaysian inference/OCR/recovery, T §14	Architecture cannot meet the residency promise yet.	Actual service/data-path and capacity validation.
Managed distribution eligibility, T §§6/7	Deployment assumptions may fail.	Existing Apple account/eligibility question, scheduled by Hafiz.
Official formats/connectors, D §8/§11	Some official re-entry remains separate.	Approved receiving format and interface evidence.
Commercial assumptions, H D10	No approved affordability, staffing or payment position.	Partner reconciles model; Hafiz approves actual offer.

The research backlog also contains unsettled APDM duplication statistics, outdated workload numbers, policy/circular copies, the RPM literacy-baseline inconsistency, exact assessment/reporting rules and national-programme chronology. Do not select the most convenient number or treat an inaccessible page as proof of absence. These are preserved in [Research §10](#), [Technical §14/15](#) and [Data §10](#), not silently cleared here.

Four existing human questions remain intact: who authorises the data-sharing/controller arrangement; which APDM/SPPB formats and idMe federation are permitted; whether SiPKPM flags/signals can be exchanged; and obtaining the current BPK templates and relevant finance/remedial/special-education/language guidelines. See [Data §11](#). This list is a record, not authority to contact KPM.

A remaining source tension concerns behaviour/wellbeing features. P lists cautious Later options, while D excludes behaviour/SEL scoring. Carry them as research possibilities only: implementing collection requires a new explicit scope/governance decision. Do not interpret “Later” as advance permission to collect excluded data.

16. Understand how to use, update and share the truth

The evidence foundation is not a single undifferentiated file. H records owner decisions and corrections; P governs product tiers; T explains proposed architecture; D governs data; R preserves research; the partner owns figures; this sourcebook explains their relationship. New accepted facts should update the appropriate master and then the synthesis, not silently create a competing specification.

For each material claim, record its precise wording, direct source/section, date, status, permitted audience, limitations and use in the book. Keep D/G decisions and earlier P01–P12 proposition IDs intact. A proposition ID is not an experimental result. Local source paths belong in internal editions, not in public or controlled external HTML.

The publication flow is **source** → **qualified claim** → **approved audience wording** → **generated edition** → **content/visual/disclosure checks** → **owner release decision**. Hidden internal sections are still present in a file. The supplied partner share builder was shown to retain restricted paragraphs and script-held dummy content; it is not a confidentiality guarantee. Build only the intended reader payload and inspect its assets/scripts as well as visible pages.

Update the Markdown master, then regenerate HTML/PDF/DOCX. Preserve a manifest linking sources/renderers/outputs, a dated change log and the exact checks performed. Recheck volatile sources before circulation or a commitment. A green renderer is not proof of readability; a content review is not software QA; owner approval of direction is not counterpart consent.

The six-part reading structure was approved by Hafiz with “Yes” after the external draft handback. This authorises local internal drafting and checks, not publication, commercial figure duplication, code deployment or changes to Claude's deck. Final edition/review status is recorded separately in the acceptance note.

Appendix A. Complete feature index

This is a compact row-by-row index of P §§3–8. Repeated features retain both source locations rather than becoming new scope. C = Core; F = Fast-follow; L = Later; B = same-launch guided beta. All remain proposed. Each profile's input, human authority and validation are in chapter 5; mechanics in 4/7/8.

Pillar	Features / tier
1: learning evidence	Formative check-ins C; MCQ marking C; handwritten fill-in/numeric capture F; standalone attendance C; APDM sync F conditional; teacher-maintained mastery C; class dashboard C; essay assistance L; early-warning flags L.
2: classroom	Live polls/quizzes C; simple game templates F; textbook-linked interactive delivery F conditional; supported offline resources/register/local marking C.
3: preparation	AI worksheet/quiz C; template RPH C; full AI-RPH shared backend B; textbook-grounded worksheet F conditional; reading-level adaptation C; BM/English/Mandarin/Tamil parity C; student-specific differentiated material F; rubric generation F.
4: teaching support	Student strengths/weakness summaries C; rule-based remediation C; AI learning-plan narrative B; teacher chat assistant B; rule-based class pattern detection C; autonomous teacher-free adaptive path excluded from MVP/v1 direction.
5: efficiency	Teacher-approved parent communication templates C; AI-written admin drafts B; APDM-aware attendance F conditional (same integration as pillar 1); RPH library/sharing C; handwritten grading capture F conditional (same capture as pillar 1); SEL/behaviour L research only, data exclusion still controls.
6: family	Curated progress C; attendance/announcements C; age-gated visibility C; periodic digest C; direct messaging F; full raw gradebook/history L subject to need/governance.
Cross-cutting	First-week onboarding/coaching, champions, four product languages and accessibility C; no Phase-1 KPM integration; teacher-only supplied devices; Malaysia-resident identifiable lifecycle; teacher decision authority.

Appendix B. Field/event and authority dictionary

The six record families in chapter 8 map respectively to D §§5.1–5.6. Lesson and item records preserve source/curriculum and generation versions; longitudinal student records add teacher judgement, interventions and controller-authorised school transfer; class/school records aggregate permitted evidence; teacher records add prompt category, model usage and draft edits; family records add notice, access and complaint state. Chapter 8's access matrix and lifecycle apply to each family. Retention is not one universal period: the controller must set class-specific periods, including shorter raw prompt/telemetry storage and separate access-log retention. Field presence does not authorise collection; approved purpose, observable provenance and permission remain prerequisites.

Concept	Meaning and distinction	Detailed source
Curriculum/version ID	Identifies the exact standard context; changing versions must not silently rewrite past meaning.	T §4; D §8
Lesson / worksheet instance	A particular planned/delivered session or assigned set, not just a reusable template.	D §§5.1/5.2
Response / attempt	What was observed in a channel, with student/item/mode and recorded evidence.	D §5.2
Target TP / estimate / formal TP	Item intent, computed interpretation and teacher's official judgement are different concepts.	D §5 boundary; P §9
Override	Teacher correction with reason and audit, not an unexplained overwrite.	D §5.2
Planned / taught	Teacher plan versus confirmed actual activity, not inferred from opening a file.	D §5.1
Notice / grant / objection	Communication, permission and complaint states, not interchangeable checkboxes.	D §§5.6/7.5
Event / receipt time	When work was recorded versus when a disconnected device later synchronises.	D §5; implementation detail to validate
Aggregate / pseudonym	Group summary versus potentially re-identifiable individual data; different safeguards apply.	D §§7.3/7.4/7.8

Appendix C. Evidence and claim map

Claim group	Status	Source / date / use
Proposal and roles	Owner-approved proposal, not agreement	H §7; REC-01–REC-08, September 2026; chapters 1/3/10/13/16
Product and data	Proposed design	P/D, reviewed 6 September 2026; chapters 4–12/14, appendices A/B
Existing platform patterns	Dated internal audit	P §1, 6 September 2026; chapter 6; no current code re-audit
Research lessons and gaps	Dated synthesis with limits	R/T/D, 6 September 2026; chapters 2/7/15; no uncited headline statistics
Kota Buku remit	Publicly stated	Primary About page, checked 7 September 2026; chapter 1 only
Guided preparation evidence	Dated study finding	EEF/NFER report December 2024, checked 7 September 2026; chapter 2, no transferred effect
Grounded MCQ study	Narrow preprint finding	arXiv v1, 6 August 2025, checked 7 September 2026; chapter 7, no human-validation claim
Standards-exchange / accessibility	Reference/target, not certification	CASE and WCAG primary pages, checked 7 September 2026; chapters 7/9
Retry, escalation and evaluation examples	Proposed acceptance elaboration	Derived from P/T/D and REC-05; chapters 6/9–12; not executed contracts or tests

The companion internal evidence register records exact verification limits and edition checks. Any later external quotation requires its own audience-safe wording and refreshed source check.

Appendix D. Decision and assumption history

Governing index: [Handoff §7](#) and dated [§13](#) corrections for D1–D11/C4; [reconciliation §2](#) for REC-01–REC-08. The chapter 15 items are open as of 7 September 2026. Role/support items are owner-approved proposals with counterpart acceptance unconfirmed; legal/rights/age/format items are unresolved authority or policy checks; curriculum/capture/residency/distribution items are proposed requirements awaiting validation; commercial assumptions remain partner-controlled and unaccepted. Source specifications were reviewed on 6 September; REC approvals are dated 7 September. None is classified as an established operating fact. This index supplements rather than replaces the source verification flags.

The sequence matters: initial research → independent corrections → partner-pack reconciliation with D8/D9/D10 → blended C4 → benefit-first visual direction → framework reconciliation → separate external/internal blueprints → external draft → internal structure approval. Older instructions can remain historically accurate while no longer controlling the next action.

Do not treat the partner review's old “C4 open” or branding/device assumptions as current. Do not treat the early review-only “no drafting” boundary as current after explicit document approval. Conversely, do not treat approval to draft as permission to publish or implement. The current Handoff entry, REC record and acceptance note identify the active boundary.

Appendix E. Acceptance catalogue

Representative check	Required evidence before claiming delivery
Prepare material	Correct source/objective, valid answer, teacher edit/reject path and language suitability.
Capture paper responses	Correct student/item match, ambiguous marks resolved, no invented timing, correction audit.
Work offline/retry	Local persistence, understandable pending state, no duplicate attempt after reconnect, conflict behaviour tested.
Advise teacher	Evidence traceable, missing data distinct, teacher chooses action, no automated formal TP.
Share with family	Correct recipient, age policy, own-record isolation, teacher approval and periodic visibility.
Support and incident	Safe escalation evidence, accountable route, recovery and teacher task restored.
Release full-RPH / analysis / assistant / admin trial	Shared service, selected-teacher access, teaching-quality review and controlled wider release.
Publish a document	Source/edition freshness, no hidden restricted payload, readable language/figures and owner approval of actual file.

Thresholds, fixtures and named delivery owners are not invented here. The catalogue is a requirement for future evidence, not a declaration tests passed.

Appendix F. Glossary

Term	Plain meaning
RPH / RPT	Daily lesson plan / yearly teaching plan.
DSKP / TP	Curriculum and assessment standards / teacher-determined mastery level.
Grounding / retrieval	Selecting relevant permitted source material to guide a generated draft.
Deterministic check	A defined rule such as matching an answer key, not generated judgement.
Inference / OCR	Running a model to produce an output / recognising text from an image.
Controller / processor	Party determining permitted purposes / party processing on its instructions.
Idempotent retry	Repeating a submission does not create duplicate effects.
Core / guided beta	Proposed general-launch functions / selected-teacher supervised full-RPH, AI-analysis, assistant and AI-admin use at that same launch.
Fast-follow / Later	Subsequent tier after prerequisites / more advanced conditional development.
Evidence debt	A claim or requirement still lacking the needed verification.

Appendix G. Source provenance and safe-reading index

Read [research findings](#) for the questions and landscape, [product specification](#) for exact tiers, [technical blueprint](#) for architecture/validation and [data specification](#) for fields/governance. Read [Handoff](#) for decisions and source-qualified corrections. [Reconciliation](#) explains the partner-framework dispositions.

The partner files are read-only, controlled internal inputs; they contain superseded assumptions and restricted challenge material. The downloaded framework is methodology, not new deal facts. Preserved PDFs/text in [sources](#) / retain their source dates and limits; possession does not establish licences or current applicability. The external document is a separately qualified reader edition, not this sourcebook with sections merely hidden.